

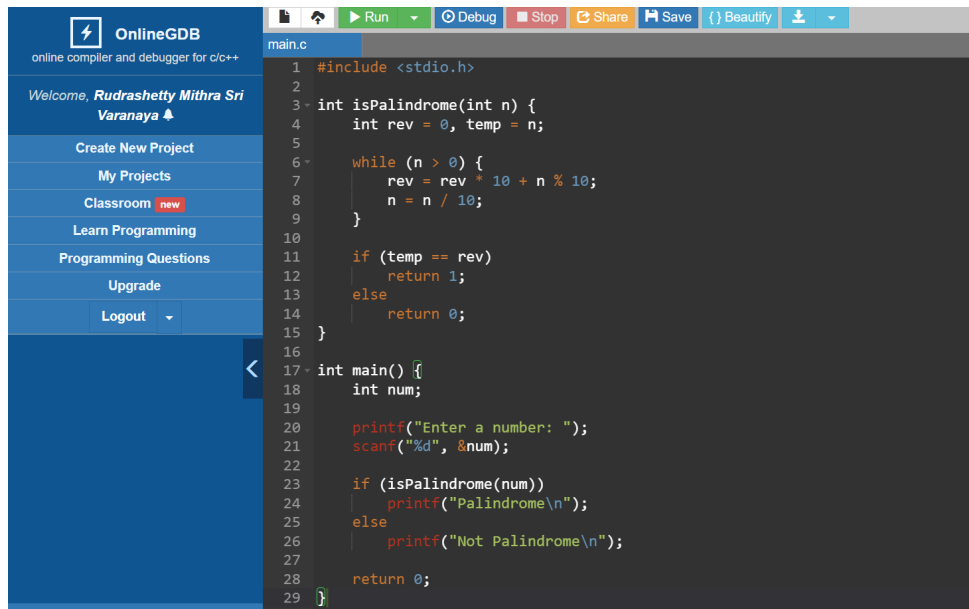
BITS PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI

COURSE	: COMPUTER PROGRAMMING (CS F111)
COMPONENT	: Practical Week 8: User defined functions

1) Write a C program using a function to check whether a given number is palindrome or not.

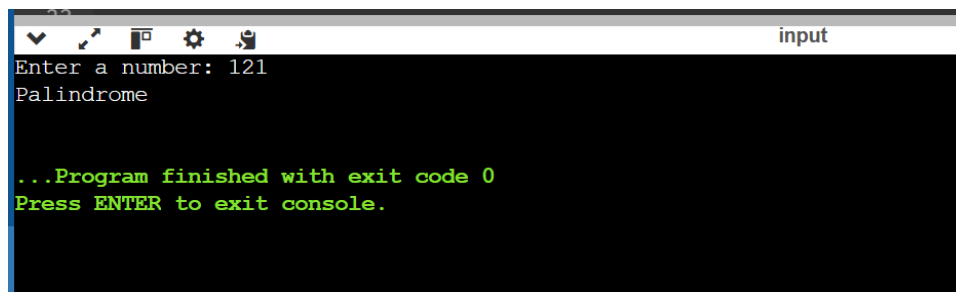
Answer:

Code:

The screenshot shows the OnlineGDB interface. On the left is a sidebar with navigation links: 'Welcome, Rudrashetty Mithra Sri Varanaya', 'Create New Project', 'My Projects', 'Classroom', 'Learn Programming', 'Programming Questions', 'Upgrade', and 'Logout'. The main area displays a C program in 'main.c'. The code defines a function 'isPalindrome' that reverses the digits of a number and compares it with the original. The 'main' function prompts the user to enter a number and prints whether it is a palindrome or not. The code is as follows:

```
1 #include <stdio.h>
2
3 int isPalindrome(int n) {
4     int rev = 0, temp = n;
5
6     while (n > 0) {
7         rev = rev * 10 + n % 10;
8         n = n / 10;
9     }
10
11     if (temp == rev)
12         return 1;
13     else
14         return 0;
15 }
16
17 int main() {
18     int num;
19
20     printf("Enter a number: ");
21     scanf("%d", &num);
22
23     if (isPalindrome(num))
24         printf("Palindrome\n");
25     else
26         printf("Not Palindrome\n");
27
28     return 0;
29 }
```

Input and Output:

The screenshot shows a terminal window titled 'input'. It displays the execution of the C program. The user enters '121' when prompted 'Enter a number:'. The program outputs 'Palindrome'. Below this, a green message states '...Program finished with exit code 0' and 'Press ENTER to exit console.'.

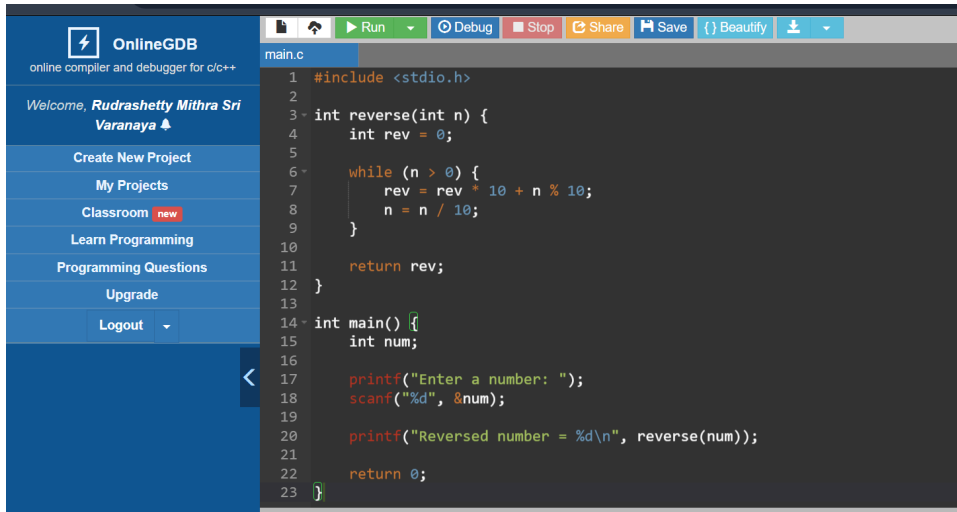
```
input
Enter a number: 121
Palindrome

...Program finished with exit code 0
Press ENTER to exit console.
```

2) Write a C program using a function to reverse a given number.

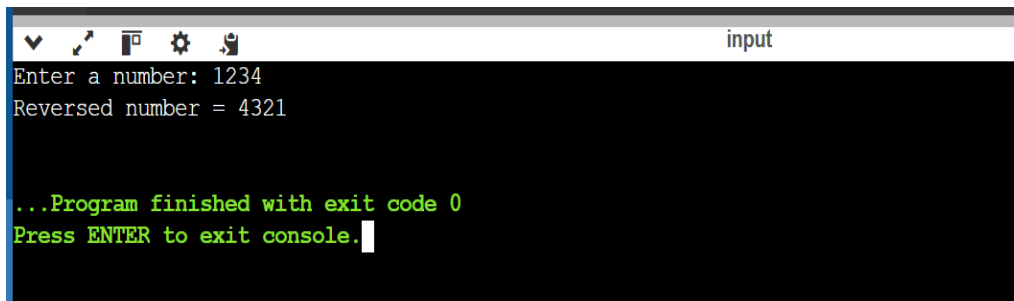
Answer:

Code:

The screenshot shows the OnlineGDB interface. On the left is a sidebar with navigation links: 'Welcome, Rudrashetty Mithra Sri Varanaya', 'Create New Project', 'My Projects', 'Classroom' (marked as new), 'Learn Programming', 'Programming Questions', 'Upgrade', and 'Logout'. The main area displays a C program in a dark-themed editor. The code defines a 'reverse' function that uses a while loop to calculate the reverse of a number by repeatedly taking the last digit and shifting it to the front. The 'main' function prompts the user to enter a number, reads it, and prints the reversed number using the 'reverse' function.

```
1 #include <stdio.h>
2
3 int reverse(int n) {
4     int rev = 0;
5
6     while (n > 0) {
7         rev = rev * 10 + n % 10;
8         n = n / 10;
9     }
10
11     return rev;
12 }
13
14 int main() {
15     int num;
16
17     printf("Enter a number: ");
18     scanf("%d", &num);
19
20     printf("Reversed number = %d\n", reverse(num));
21
22     return 0;
23 }
```

Input and Output:

The screenshot shows the program's execution output in a terminal window titled 'input'. It displays the prompt 'Enter a number: 1234', followed by the output 'Reversed number = 4321'. At the bottom, it shows the message '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a cursor.

```
input
Enter a number: 1234
Reversed number = 4321

...Program finished with exit code 0
Press ENTER to exit console.
```

3) Write a C program to pass an array to a function and find the largest element in the array.

Answer:

Code: Scroll Down,

The screenshot shows the OnlineGDB web interface. On the left is a sidebar with navigation links: 'Create New Project', 'My Projects', 'Classroom' (marked as new), 'Learn Programming', 'Programming Questions', 'Upgrade', and 'Logout'. The main area displays a C program named 'main.c'. The code defines a function 'findLargest' that takes an array and its size, iterates through the array to find the maximum value, and returns it. The 'main' function prompts the user for the number of elements and the elements themselves, then calls 'findLargest' and prints the result.

```
1 #include <stdio.h>
2
3 int findLargest(int arr[], int n) {
4     int max = arr[0];
5
6     for (int i = 1; i < n; i++) {
7         if (arr[i] > max)
8             max = arr[i];
9     }
10
11     return max;
12 }
13
14 int main() {
15     int n;
16
17     printf("Enter number of elements: ");
18     scanf("%d", &n);
19
20     int arr[n];
21
22     printf("Enter elements:\n");
23     for (int i = 0; i < n; i++) {
24         scanf("%d", &arr[i]);
25     }
26
27     printf("Largest element = %d\n", findLargest(arr, n));
28
29     return 0;
30 }
```

Input and Output:

The screenshot shows a terminal window titled 'input'. It displays the program's execution: the user enters '5' for the number of elements and '10 18 46 69 100' for the elements. The program outputs 'Largest element = 100'. At the bottom, it shows '...Program finished with exit code 0' and 'Press ENTER to exit console.'.

```
Enter number of elements: 5 10 18 46 69 100
Enter elements:
Largest element = 100

...Program finished with exit code 0
Press ENTER to exit console.
```

4)Write a C program using a function to print the multiplication table of a given number.

Answer:

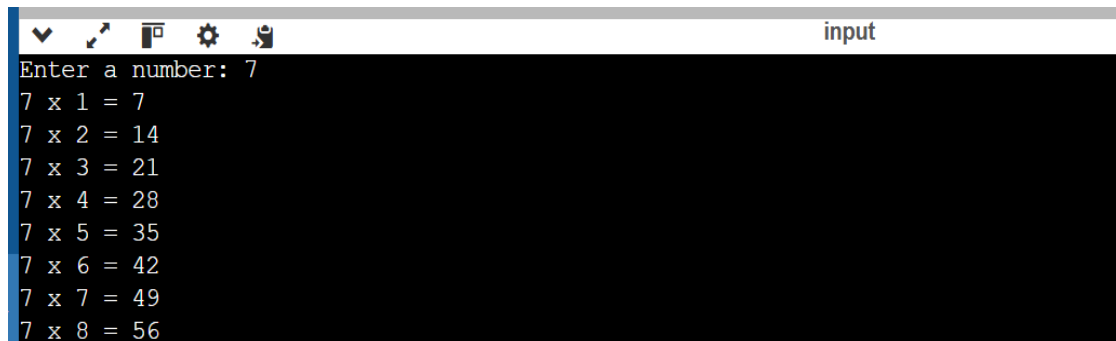
Code: Scroll down,



The screenshot shows the OnlineGDB web interface. On the left is a sidebar with the OnlineGDB logo, a welcome message for 'Rudrashetty Mithra Sri Varanaya', and navigation links: 'Create New Project', 'My Projects', 'Classroom' (with a 'new' badge), 'Learn Programming', 'Programming Questions', 'Upgrade', and a 'Logout' button. The top of the editor has a toolbar with icons for file operations, a 'Run' button, 'Debug', 'Stop', 'Share', 'Save', 'Beautify', and a download icon. The main editor area displays a C program named 'main.c' with the following code:

```
1 #include <stdio.h>
2
3 void printTable(int n) {
4     for (int i = 1; i <= 10; i++) {
5         printf("%d x %d = %d\n", n, i, n * i);
6     }
7 }
8
9 int main() {
10     int num;
11
12     printf("Enter a number: ");
13     scanf("%d", &num);
14
15     printTable(num);
16
17     return 0;
18 }
```

Input and Output:



The screenshot shows the 'input' window of the OnlineGDB interface. It contains the input and output of the program. The input is '7', and the output is a multiplication table for the number 7, showing products from 7 x 1 to 7 x 8.

```
Enter a number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
```